MH4110 Partial Differential Equations

Handout 8 - Green's Identities and Green's Functions

Outline

The Green's function is a kind of universal solution for harmonic functions in a domain. All other harmonic functions can be expressed in terms of it.

- Green's First Identity
- Green's Second Identity
- Green's Functions
- Half-Space and Sphere

§7.1 Green's First Identity

Diffusions

The solution to the diffusion (heat) initial-value problem on the whole line

$$\begin{aligned} u_t &= ku_{xx}, \quad -\infty < x < \infty, \\ u(x,0) &= \phi(x), \quad -\infty < x < \infty, \end{aligned} \tag{1}$$

is given by the formula

$$u(x,t) = S(x,t)*\phi(x) = \int_{-\infty}^{\infty} S(x-y,t)\phi(y)dy = \frac{1}{\sqrt{4\pi kt}}\int_{-\infty}^{\infty} \exp\Big(-\frac{(x-y)^2}{4kt}\Big)\phi(y)dy. \tag{2}$$

The fundamental solution or Gaussian kernel

$$S(x,t) = \frac{1}{\sqrt{4\pi kt}}\exp\Big(-\frac{x^2}{4kt}\Big) \tag{3}$$

has the Dirac delta function $\delta(x)$ as its initial data.

§7.1 Green's First Identity

Some notations

$$\nabla u = \operatorname{grad} u = u_x \mathbf{i} + u_y \mathbf{j} + u_z \mathbf{k} = (u_x, u_y, u_z); \tag{4}$$

$$\operatorname{div} \mathbf{F} = \nabla \cdot \mathbf{F} = u_x + v_y + w_z, \tag{5}$$

where $\mathbf{F} = (u, v, w)$, and

$$\begin{aligned} \nabla \times \mathbf{F} = \operatorname{curl} \mathbf{F} &= \begin{vmatrix} \mathrm{i} & \mathrm{j} & \mathrm{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ u & v & w \end{vmatrix} \\ &= (w_y - v_z)\mathbf{i} + (u_z - w_x)\mathbf{j} + (v_x - u_y)\mathbf{k}. \end{aligned} \tag{6}$$

Notice that

$$\Delta u = u_{xx} + u_{yy} + u_{zz} = \operatorname{div}(\nabla u). \tag{7}$$

$$|\nabla u|^2 = |\operatorname{grad} u|^2 = u_x^2 + u_y^2 + u_z^2. \tag{8}$$

§7.1 Green's First Identity

We will write almost everything in this chapter for the three-dimensional case. (However, using two dimensions is okay, too, even n dimensions.) Thus we write

$$\iiint_D \cdots d\mathbf{x} = \iiint_D \cdots dx\, dy\, dz \tag{9}$$

if D is a three-dimensional region (a solid), and

$$\iint_{\mathrm{bdy}\, D} \cdots dS = \iint_S \cdots dS, \tag{10}$$

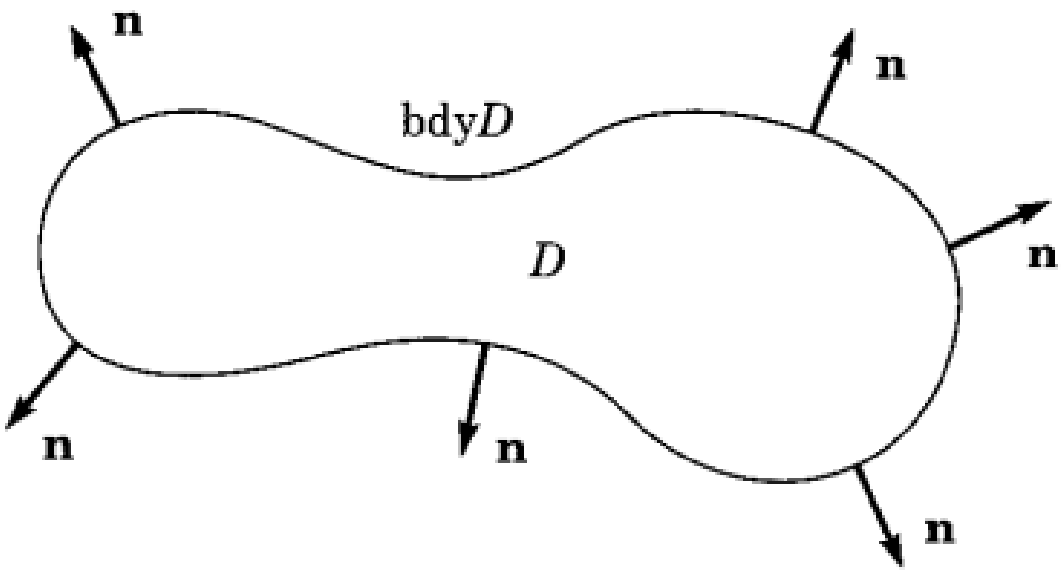
where $S = \mathrm{bdy}\, D$ is the bounding surface for the solid region D. Here dS indicates the usual surface integral, as in the calculus.

§7.1 Green's First Identity

Our basic tool in this chapter will be the divergence theorem

$$\iiint_D \operatorname{div} \mathbf{F} d\mathbf{x} = \iint_{\operatorname{bdy} D} \mathbf{F} \cdot \mathbf{n} dS, \tag{11}$$

where $\mathbf{F}$ is any vector function, D is a bounded solid region, and $\mathbf{n}$ is the unit outer normal on $\operatorname{bdy} D$



§7.1 Green's First Identity

Exercise

Show that

$$\nabla \cdot (v \nabla u) = \nabla v \cdot \nabla u + v \Delta u.$$

§7.1 Green's First Identity

Integrate

$$\nabla \cdot (v \nabla u) = \nabla v \cdot \nabla u + v \Delta u.$$

and use the divergence theorem on the left side to get

Green's first identity

$$\iint_{\text{bdy } D} v \frac{\partial u}{\partial n} dS = \iiint_D \nabla v \cdot \nabla u d\mathbf{x} + \iiint_D v \Delta u d\mathbf{x}, \tag{12}$$

where $\frac{\partial u}{\partial n} = \mathbf{n} \cdot \nabla u$ is the directional derivative in the outward normal direction. This is *Green's first identity*. It is valid for any solid region D and any pair of functions u and v.

For example, we could take $v \equiv 1$ to get

$$\iint_{\text{bdy } D} \frac{\partial u}{\partial n} dS = \iiint_D \Delta u d\mathbf{x}. \tag{13}$$

§7.1 Green's First Identity

Consider the Neumann problem in any domain D

$$\begin{cases} \Delta u = f(\boldsymbol{x}), & \text{in } D, \\ \frac{\partial u}{\partial n} = h(\boldsymbol{x}), & \text{on bdy } D. \end{cases} \tag{14}$$

By (13) we have

$$\iint\limits_{\text{bdy } D} h \, dS = \iiint\limits_{D} f \, d\mathbf{x}. \tag{15}$$

It follows that the data (f and h) are not arbitrary but are required to satisfy condition (15). Otherwise, there is no solution. In that sense the Neumann problem (14) is not completely well-posed.

§7.1 Green's First Identity

$$u(r,\theta) = (a^2 - r^2)\int_0^{2\pi} \frac{h(\phi)}{a^2 - 2ar\cos(\theta-\phi) + r^2}\frac{d\phi}{2\pi}. \tag{16}$$

This single formula (16), known as Poisson's formula. It expresses any harmonic function inside a circle in terms of its boundary values.

Mean Value Property

In three dimensions the mean value property states that the average value of any harmonic function over any sphere equals its value at the center.

Proof: Let D be a ball, $\{|\boldsymbol{x}| < a\}$, say; that is, $\left\{x^2 + y^2 + z^2 < a^2\right\}$. Then $\mathrm{bdy}\ D$ is the sphere (surface) $\{|\boldsymbol{x}| = a\}$. Let $\Delta u = 0$ in any region that contains D and $\mathrm{bdy}\ D$. For a sphere, $\boldsymbol{n}$ points directly away from the origin, so that

$$\frac{\partial u}{\partial n} = \boldsymbol{n}\cdot\nabla u = \frac{\boldsymbol{x}}{r}\cdot\nabla u = \frac{x}{r}u_x + \frac{y}{r}u_y + \frac{z}{r}u_z = \frac{\partial u}{\partial r},$$

where $r = \sqrt{x^2 + y^2 + z^2} = |\boldsymbol{x}|$ is the spherical coordinate, the distance of the point (x, y, z) from the center $\mathbf{0}$ of the sphere.

§7.1 Green's First Identity

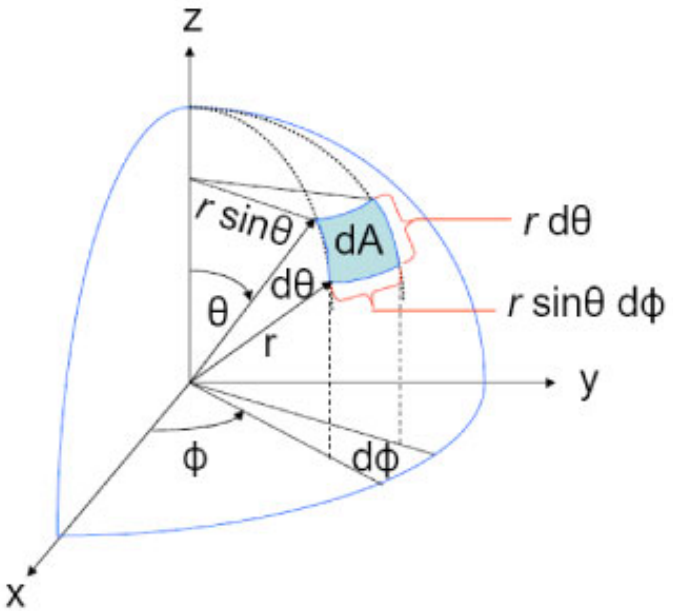
Proof (cont'd): Therefore, (13) becomes

$$\iint\limits_{\text{bdy } D} \frac{\partial u}{\partial r} dS = 0. \tag{17}$$

Let's write this integral in spherical coordinates, (r, θ, ϕ). Explicitly, (17) takes the form

$$\int_0^{2\pi} \int_0^{\pi} u_r(a,\, \theta,\, \phi) a^2 \sin\theta d\theta d\phi = 0$$

since $r = a$ on $\text{bdy } D$.



§7.1 Green's First Identity

Proof (cont'd): We divide this by the constant $4\pi a^2$ (the area of $\mathrm{bdy}\ D$). This result is valid for all $a > 0$, so that we can think of a as a variable and call it r. Then we pull $\partial/\partial r$ outside the integral, obtaining

$$\frac{\partial}{\partial r}\left[\frac{1}{4\pi}\int_0^{2\pi}\int_0^{\pi} u(r,\,\theta,\,\phi)\sin\theta d\theta d\phi\right] = 0.$$

Thus

$$\frac{1}{4\pi}\int_0^{2\pi}\int_0^{\pi} u(r,\,\theta,\,\phi)\sin\theta d\theta d\phi$$

is independent of r. This expression is precisely the average value of u on the sphere $\{|\boldsymbol{x}| = r\}$. In particular, if we let $r \to 0$, we get

$$\frac{1}{4\pi}\int_0^{2\pi}\int_0^{\pi} u(\mathbf{0})\sin\theta d\theta d\phi = u(\mathbf{0}).$$

That is

$$\frac{1}{4\pi}\int_0^{2\pi}\int_0^{\pi} u(r,\,\theta,\,\phi)\sin\theta d\theta d\phi = u(\mathbf{0})$$

or

$$\frac{1}{\text{area of } S}\iint_S u\, dS = u(\mathbf{0}). \tag{18}$$

This proves the mean value property in three dimensions.

§7.1 Green's First Identity

Exactly as in two dimensions, we deduce the maximum principle from the mean value property (Proof omitted).

Maximum Principle

If D is any solid region, a nonconstant harmonic function in D cannot take its maximum value inside D, but only on $\mathrm{bdy}D$.

Uniqueness of Dirichlet's Problem

$$\begin{cases} \Delta u = f, & \text{in } D, \\ u = h & \text{on } \mathrm{bdy}(D). \end{cases} \tag{19}$$

We gave one proof in Section 6.1 using the maximum principle. Now we give another proof by the energy method. If we have two harmonic functions u_1 and u_2 with the same boundary data, then their difference $u = u_1 - u_2$ is harmonic and has zero boundary data. We go back to Green's first identity

$$\iint_{\mathrm{bdy}\, D} v \frac{\partial u}{\partial n} dS = \iiint_D \nabla v \cdot \nabla u d\mathbf{x} + \iiint_D v \Delta u d\mathbf{x}$$

and substitute $v = u$.

Uniqueness of Dirichlet's Problem (cont'd)

Since u is harmonic, we have $\Delta u = 0$ and

$$\iint\limits_{\text{bdy } D} u \frac{\partial u}{\partial n} dS = \iiint\limits_{D} |\nabla u|^2 d\mathbf{x}. \tag{20}$$

Since $u = 0$ on $\text{bdy} D$, the left side of (20) vanishes. Therefore,

$$\iiint\limits_{D} |\nabla u|^2 d\mathbf{x} = 0.$$

It follows that $|\nabla u|^2 \equiv 0$ in D.

Now a function with vanishing gradient must be a constant (provided that D is connected). So $u(\boldsymbol{x}) \equiv C$ throughout D. But u vanishes somewhere (on bdy D), so C must be 0. This proves the uniqueness of the Dirichlet problem.

Dirichlet's Principle

Among all the functions $w(\boldsymbol{x})$ in D that satisfy the Dirichlet boundary condition

$$w = h(\boldsymbol{x}) \text{ on } \mathrm{bdy}D, \tag{21}$$

the lowest energy occurs for the harmonic function satisfying (21).

This is an important mathematical theorem based on the physical idea of energy. In the present context the energy is defined as

$$E[w] = \frac{1}{2}\iiint_D |\nabla w|^2 d\mathbf{x}. \tag{22}$$

This is the pure potential energy, there being no kinetic energy because there is no motion. Now it is a general principle in physics that any system prefers to go to the state of lowest energy, called the ground state. Thus the harmonic function is the preferred physical stationary state.

Mathematically, Dirichlet's principle can be stated precisely as follows:

Let $u(\boldsymbol{x})$ be the unique harmonic function in D that satisfies (21). Let $w(\boldsymbol{x})$ be any function in D that satisfies (21). Then

$$E[w] \geq E[u]. \tag{23}$$

Dirichlet's Principle

Let $u(\boldsymbol{x})$ be the unique harmonic function in D that satisfies (21). Let $w(\boldsymbol{x})$ be any function in D that satisfies (21). Then

$$E[w] \geq E[u].$$

Proof: To prove Dirichlet's principle, we let $v = u - w$ and expand the square in the integral

$$E[w] = \frac{1}{2}\iiint_D |\nabla(u-v)|^2 d\boldsymbol{x} = E[u] - \iiint_D \nabla u \cdot \nabla v d\boldsymbol{x} + E[v]. \tag{24}$$

Next we apply Green's first identity

$$\iint_{\text{bdy } D} v\frac{\partial u}{\partial n} dS = \iiint_D \nabla v \cdot \nabla u d\mathbf{x} + \iiint_D v\Delta u d\mathbf{x}$$

to the pair of functions u and v. In the Green's first identity, two of the three terms are zero because $v = 0$ on bdy D and $\Delta u = 0$ in D. Therefore, the middle term in (24) is also zero. Thus

$$E[w] = E[u] + E[v].$$

Since it is obvious that $E[v] \geq 0$, we deduce that $E[w] \geq E[u]$. This means that the energy is smallest when $w = u$. This proves Dirichlet's principle.

§7.2 Green's Second Identity

Green's Second Identity

$$\iiint_D (u\Delta v - v\Delta u)d\mathbf{x} = \iint_{\text{bdy } D} \left(u\frac{\partial v}{\partial n} - v\frac{\partial u}{\partial n}\right) dS \tag{25}$$

Just like Green's First Identity, it is valid for any pair of functions u and v.

Exercise

Deduce Green's Second Identity.

§7.2 Green's Second Identity

Exercise

Let

$$v(\boldsymbol{x}) = -\frac{1}{4\pi|\boldsymbol{x} - \boldsymbol{x}_0|}.$$

Show that $\Delta v = 0$.

§7.2 Green's Second Identity

Representation Formula

If $\Delta u = 0$ in D, then

$$u(\boldsymbol{x}_0) = \iint\limits_{\text{bdy } D} \left(-u(\boldsymbol{x}) \frac{\partial}{\partial n} \left(\frac{1}{|\boldsymbol{x} - \boldsymbol{x}_0|} \right) + \frac{1}{|\boldsymbol{x} - \boldsymbol{x}_0|} \frac{\partial u}{\partial n} \right) \frac{dS}{4\pi} \tag{26}$$

Proof: The representation formula (26) is the special case of Green's Second Identity with the choice

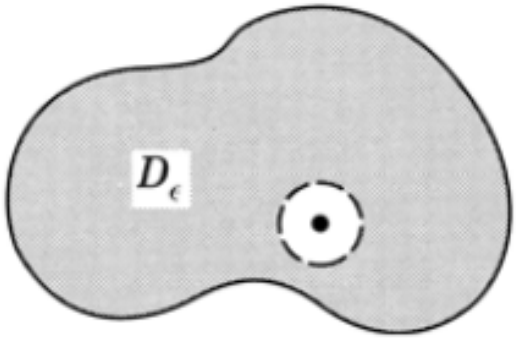
$$v(\boldsymbol{x}) = -\frac{1}{4\pi|\boldsymbol{x} - \boldsymbol{x}_0|}.$$

Clearly, the right side of Green's Second Identity agrees with (26). Also, $\Delta u = 0$ and $\Delta v = 0$, which kills the left side of Green's Second Identity. So where does the left side of (26) come from?

From the fact that the function $v(\boldsymbol{x})$ is infinite at the point $\boldsymbol{x}_0$. Therefore, it is forbidden to apply Green's Second Identity in the whole domain D.

§7.2 Green's Second Identity

Proof (Cont'd): So let's take a pair of scissors and cut out a small ball around $\boldsymbol{x}_0$. Let D_ϵ be the region D with this ball (of radius ϵ and center $\boldsymbol{x}_0$) excised.



For simplicity let $\boldsymbol{x}_0$ be the origin. Then $v(\boldsymbol{x}) = -1/(4\pi r)$, where $r = \sqrt{x^2 + y^2 + z^2} = |\boldsymbol{x}|$. Writing down Green's Second Identity with this choice of v, we have, since $\Delta u = 0 = \Delta v$ in D_ϵ,

$$-\iint\limits_{\text{bdy } D_\epsilon} \left(u \frac{\partial}{\partial n}\left(\frac{1}{r}\right) - \frac{1}{r}\frac{\partial u}{\partial n} \right) dS = 0.$$

But bdy D_ϵ consists of two parts: the original boundary bdy D and the sphere $\{r = \epsilon\}$. On the sphere, $\partial/\partial n = -\partial/\partial r$.

§7.2 Green's Second Identity

Proof (Cont'd): Thus the surface integral breaks into two pieces,

$$-\iint\limits_{\text{bdy } D}\left(u\frac{\partial}{\partial n}\left(\frac{1}{r}\right)-\frac{1}{r}\frac{\partial u}{\partial n}\right)dS=\iint\limits_{r=\epsilon}\left(-u\frac{\partial}{\partial r}\left(\frac{1}{r}\right)+\frac{1}{r}\frac{\partial u}{\partial r}\right)dS. \tag{27}$$

This identity (27) is valid for any small $\epsilon > 0$. Our representation formula (26) would follow provided that we could show that the right side of (27) tended to $4\pi u(\mathbf{0})$ as $\epsilon \to 0$.

Now, on the little spherical surface $\{r = \epsilon\}$, we have

$$\frac{\partial}{\partial r}\left(\frac{1}{r}\right)=-\frac{1}{r^2}=-\frac{1}{\epsilon^2},$$

so that the right side of (27) equals

$$\frac{1}{\epsilon^2}\iint\limits_{r=\epsilon}u\,dS+\frac{1}{\epsilon}\iint\limits_{r=\epsilon}\frac{\partial u}{\partial r}\,dS=4\pi\bar{u}+4\pi\epsilon\frac{\bar{\partial u}}{\partial r}, \tag{28}$$

where $\bar{u}$ denotes the average of $u(\boldsymbol{x})$ on the sphere $\{|\boldsymbol{x}| = r = \epsilon\}$, and $\frac{\bar{\partial u}}{\partial r}$ denotes the average value of $\frac{\partial u}{\partial n}$ on this sphere. As $\epsilon \to 0$, the expression (28) approaches

$$4\pi u(\mathbf{0})+4\pi\cdot 0\cdot\frac{\partial u}{\partial r}(\mathbf{0})=4\pi u(\mathbf{0})$$

because u is continuous and $\partial u/\partial r$ is bounded. Thus (27) turns into (26), and this completes the proof.

§7.2 Green's Second Identity

Exercise

Derive the representation formula for harmonic functions in two dimensions:

$$u(\boldsymbol{x}_0) = \frac{1}{2\pi} \int\limits_{\text{bdy } D} \left[u(\boldsymbol{x}) \frac{\partial}{\partial n} \left(\log |\boldsymbol{x} - \boldsymbol{x}_0| \right) - \frac{\partial u}{\partial n} \log |\boldsymbol{x} - \boldsymbol{x}_0| \right] ds$$

whenever $\Delta u = 0$ in a plane domain D and $\boldsymbol{x}_0$ is a point within D. The right side is a line integral over the boundary curve with respect to arc length. Log denotes the natural logarithm and ds the arc length on the bounding curve.

§7.3 Green's Functions

Definition

The Green's function $G(\boldsymbol{x})$ for the operator $-\Delta$ and the domain D at the point $\boldsymbol{x}_0 \in D$ is a function defined for $\boldsymbol{x} \in D$ such that:

(i) $G(\boldsymbol{x})$ possesses continuous second derivatives and $\Delta G = 0$ in D, except at the point $\boldsymbol{x} = \boldsymbol{x}_0$.

(ii) $G(\boldsymbol{x}) = 0$ for $x \in \mathrm{bdy}\ D$.

(iii) The function $G(\boldsymbol{x}) + 1/(4\pi|\boldsymbol{x} - \boldsymbol{x}_0|)$ is finite at $\boldsymbol{x}_0$ and has continuous second derivatives everywhere and is harmonic at $\boldsymbol{x}_0$.

The usual notation for the Green's function is $G(\boldsymbol{x}, \boldsymbol{x}_0)$.

It can be shown that a Green's function exists. Also, it is unique.

§7.3 Green's Functions

Theorem 1

If $G(\boldsymbol{x}, \boldsymbol{x}_0)$ is the Green's function, then the solution of the Dirichlet problem is given by the formula

$$u(\boldsymbol{x}_0) = \iint\limits_{\text{bdy } D} u(\boldsymbol{x}) \frac{\partial G(\boldsymbol{x}, \boldsymbol{x}_0)}{\partial n} dS. \tag{29}$$

The only thing wrong with this beautiful formula is that it is not usually easy to find G explicitly. Nevertheless, G can be found and thereby the Dirichlet problem can be solved for some special geometries.

Symmetry of the Green's Function

For any region D we have a Green's function $G(\boldsymbol{x}, \boldsymbol{x}_0)$. It is always symmetric:

$$G(\boldsymbol{x}, \boldsymbol{x}_0) = G(\boldsymbol{x}_0, \boldsymbol{x}) \qquad \text{for } \boldsymbol{x} \neq \boldsymbol{x}_0. \tag{30}$$

In electrostatics, $G(\boldsymbol{x}, \boldsymbol{x}_0)$ is interpreted as the electric potential inside a conducting surface $S = \mathrm{bdy}\ D$ due to a charge at a single point $\boldsymbol{x}_0$. The symmetry (30) is known as the principle of reciprocity. It asserts that a source located at the point $\boldsymbol{a}$ produces at the point $\boldsymbol{b}$ the same effect as a source at $\boldsymbol{b}$ would produce at $\boldsymbol{a}$.

§7.3 Green's Functions

The Green's function also allows us to solve Poisson's equation.

Theorem 2

The solution of the problem

$$\Delta u = f \quad \text{in } D \qquad u = h \quad \text{on bdy } D \tag{31}$$

is given by

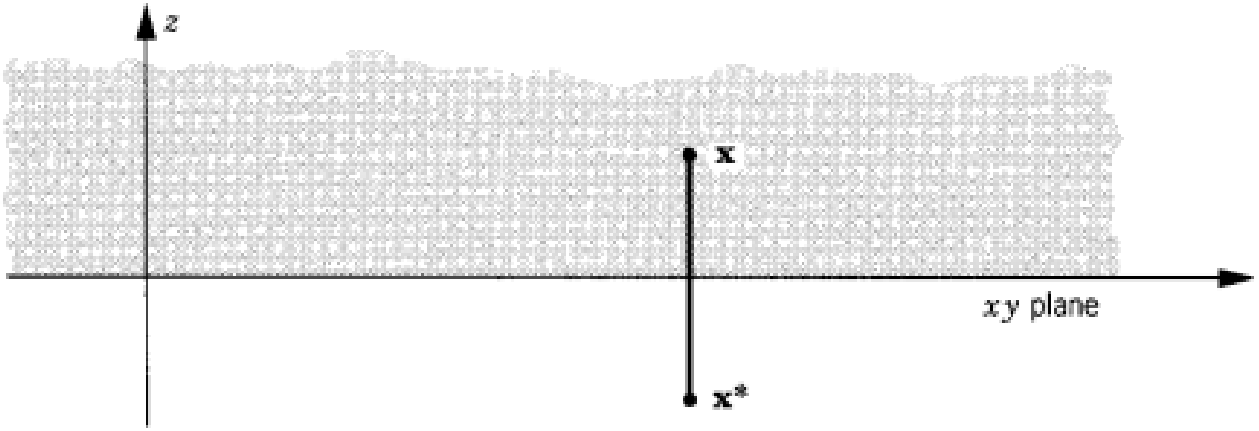
$$u(\boldsymbol{x}_0) = \iint\limits_{\text{bdy } D} h(\boldsymbol{x}) \frac{\partial G(\boldsymbol{x}, \boldsymbol{x}_0)}{\partial n} dS + \iiint\limits_{D} f(\boldsymbol{x}) G(\boldsymbol{x}, \boldsymbol{x}_0) d\boldsymbol{x}. \tag{32}$$

§7.4 Half Space

We show the Green's function for a half-space. A half-space is the region lying on one side of a plane. Although it is an infinite domain, all the ideas involving Green's functions are still valid if we impose the "boundary condition at infinity" that the functions and their derivatives tend to 0 as $|\boldsymbol{x}| \to \infty$.

We write the coordinates as $\boldsymbol{x} = (x, y, z)$. Say that the half-space is $D = \{z > 0\}$, the domain that lies above the xy plane. Each point $\boldsymbol{x} = (x, y, z)$ in D has a reflected point $\boldsymbol{x}^* = (x, y, -z)$ that is not in D. We assert that the Green's function for D is

$$G(\boldsymbol{x}, \boldsymbol{x}_0) = -\frac{1}{4\pi|\boldsymbol{x} - \boldsymbol{x}_0|} + \frac{1}{4\pi|\boldsymbol{x} - \boldsymbol{x}_0^*|}. \tag{33}$$



§7.4 Half Space

Exercise

Prove that

$$G(\boldsymbol{x}, \boldsymbol{x}_0) = -\frac{1}{4\pi|\boldsymbol{x} - \boldsymbol{x}_0|} + \frac{1}{4\pi|\boldsymbol{x} - \boldsymbol{x}_0^*|}.$$

is the Green's function for the half-space $D = \{z > 0\}$.

§7.4 Half Space

Example

Solve the Dirichlet problem

$$\Delta u = 0 \quad \text{for } z > 0, \qquad u(x, y, 0) = h(x, y).$$

Solution: We use formula (29). Notice that $\partial G/\partial n = -\partial G/\partial z|_{z=0}$ because $\boldsymbol{n}$ points downward (outward from the domain). Furthermore, on $z = 0$,

$$-\frac{\partial G}{\partial z} = \frac{1}{4\pi}\left(\frac{z + z_0}{|\boldsymbol{x} - \boldsymbol{x}_0^*|^3} - \frac{z - z_0}{|\boldsymbol{x} - \boldsymbol{x}_0|^3}\right) = \frac{1}{2\pi}\frac{z_0}{|\boldsymbol{x} - \boldsymbol{x}_0|^3}.$$

Therefore, the solution of the Dirichlet problem is

$$u(x_0, y_0, z_0) = \frac{z_0}{2\pi}\iint [(x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2]^{-3/2} h(x, y)\,dxdy,$$

where both integrals run over $(-\infty, \infty)$, noting that $z = 0$ in the integrand. In vector notation, the solution is

$$u(\boldsymbol{x}) = \frac{z_0}{2\pi}\iint\limits_{\text{bdy } D} \frac{h(\boldsymbol{x})}{|\boldsymbol{x} - \boldsymbol{x}_0|^3}\,dS.$$